

Dynamic Ex-Post Baselines for Enhancing the Credibility of Improved Forest Management Carbon Offsets

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Cite This: <https://doi.org/10.1021/acs.est.5c11387>



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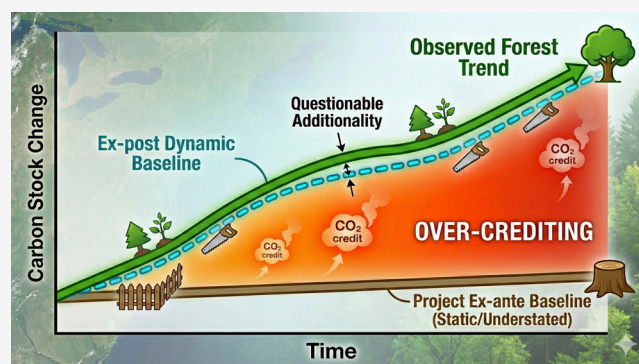
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ABSTRACT: Forest carbon offsets are pivotal to the global climate strategy, yet their claimed benefits are often undermined by flawed baselines, compromising their environmental integrity. Here, we assess five Improved Forest Management projects in China by constructing baselines using ex-post counterfactual methods based on satellite observations from 2001 to 2020. Applying methods including synthetic control, statistical matching, and machine learning, our counterfactuals, which accurately tracked pre-intervention carbon dynamics, reveal systematic underestimation of carbon sequestration in the reported baseline scenarios. We find that project forests would have continued to accumulate carbon stock naturally, contrary to the projects' static and artificially low baseline projections. Using our empirically derived baselines, we find no evidence that management activities are additional across our study samples. These biased baselines contribute to systematic over-crediting: ex-ante projected emission reductions exceed our estimates by a factor of 3.7 (uncertainty range: 1.7–5.7), while ex-post issued carbon credits exceed our estimates by a factor of 1.7 (uncertainty range: 0.8–2.5). These findings expose a fundamental limitation of current offset protocols, which fail to account for real and dynamic forest trends, leading to inflated climate benefits. We provide a scientifically grounded framework for robust, independent monitoring and dynamic baseline setting, which is crucial for ensuring the credibility of nature-based solutions.

KEYWORDS: forest carbon, baseline scenario, offset credit, voluntary carbon market



INTRODUCTION

Forests represent a critical nature-based climate solution (NCS),^{1–3} with offset projects primarily categorized into Afforestation, Reforestation, and Revegetation (ARR), Reducing Emissions from Deforestation and Degradation (REDD), and Improved Forest Management (IFM).⁴ While ARR faces land and cost constraints^{5,6} and REDD+ targets emissions from forest loss,⁷ IFM – encompassing practices like extended harvests and optimized stocking – is gaining prominence, estimated to contribute up to 16 Gt CO₂/year in negative emissions by 2030, approximately half of all NCS sequestration.^{8,9} As IFM projects become central to both voluntary and compliance carbon markets,^{10,11} persistent concerns have emerged regarding the rigor of impact assessments and the overall environmental integrity of their claimed climate benefits.^{12,13}

A central challenge in IFM projects lies in establishing credible baselines that demonstrate additionality, ensuring carbon sequestration exceeds business-as-usual scenarios. Unlike many ARR implementations that employ simplified no-forest baselines, IFM requires complex baseline construction that is highly sensitive to methodological choices. Current IFM baselines predominantly rely on historical averages,

vegetation growth models, and regional averages, creating three interconnected problems that undermine project credibility.^{14,15} First, scientific and methodological limitations plague traditional baseline approaches. Historical averages fail to capture dynamic forest growth changes driven by climate variability, market conditions, or disturbances, while vegetation models involve numerous uncertain parameters that may misrepresent site-specific conditions and management histories.¹⁶ Similarly, regional averages can disguise landscape heterogeneity and inaccurately represent individual project circumstances.¹⁷ Second, methodological inconsistencies across carbon programs enable adverse selection, where project developers strategically choose baseline methodologies, yielding the most favorable outcomes rather than the most accurate business-as-usual scenarios. Empirical evidence from California suggests potential overcrediting through advantageous regional

Received: August 17, 2025

Revised: February 3, 2026

Accepted: February 4, 2026



ACS Publications

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<https://doi.org/10.1021/acs.est.5c11387>
Environ. Sci. Technol. XXXX, XXX, XXX–XXX

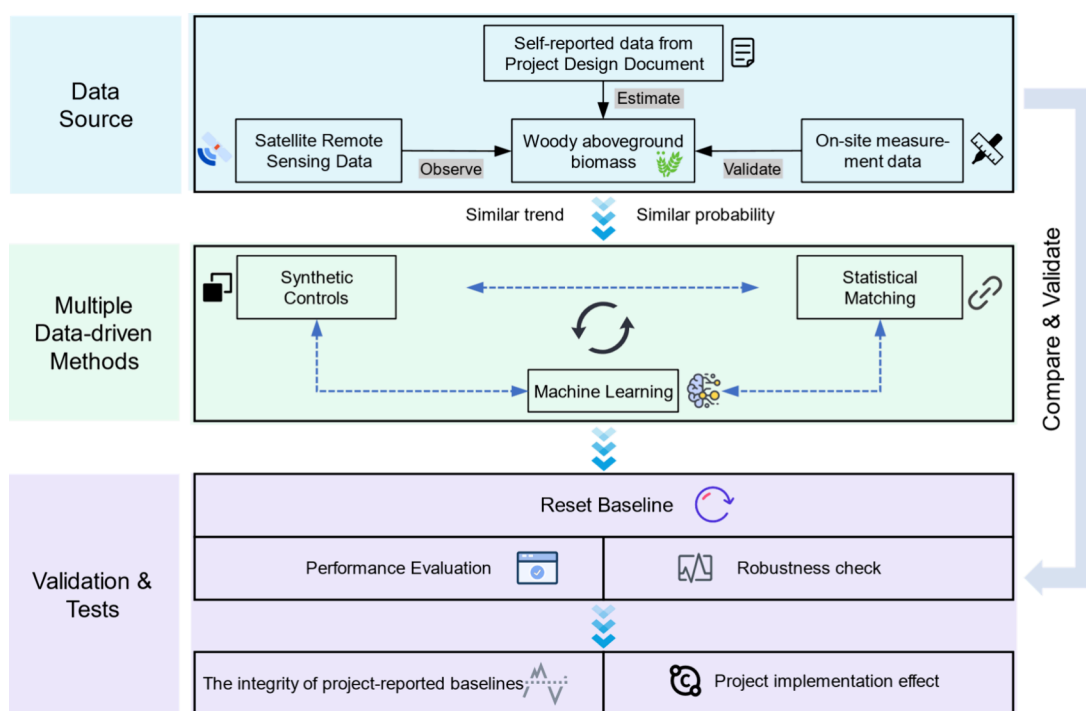


Figure 1. Research framework for dynamic ex-post baseline development of IFM projects. The framework encompasses project selection and data collection, counterfactual baseline construction integrating remote sensing data with causal inference methods, performance validation, and evaluation of project additionality and carbon credit integrity.

averages that systematically underestimate existing carbon sequestration.^{17,18} Third, monitoring deficiencies compound these issues through reliance on the baselines reported by project developers without mandatory ongoing verification of baseline greenhouse gas (GHG) removals. This information asymmetry creates incentives to underestimate baselines and grant excessive credits, mirroring patterns observed in REDD+ projects with inflated deforestation baselines.^{19–21} These limitations underscore the urgent need for more rigorous and transparent baseline setting approaches to ensure environmental integrity and market credibility in IFM carbon projects.

To address these challenges, ex-post approaches using causal inference frameworks offer a transformative solution by empirically constructing robust counterfactuals. These methods range from statistical techniques like synthetic control and matching, widely used independent evaluations of REDD+ projects to create credible counterfactuals from similar control groups,^{22–24} to machine learning algorithms that capture complex ecosystem dynamics and nonlinear relationships that traditional models cannot adequately consider.²⁵ They leverage extensive data to train predictive models that better reflect real-world forest management scenarios under varying climatic, economic, and disturbance conditions. The effectiveness of these data-driven approaches is fundamentally enabled by remote sensing integration,^{26,27} which can provide the numerous, independent, and transparent observational data required to train these models, track forest dynamics, and enable verifiable carbon accounting.²⁸ Its applications span from academic tracking of offset performance using satellite imagery^{29,30} to policy implementation, such as China's encouragement of airborne LiDAR for enhanced biomass monitoring in ARR methodologies.³¹

This study advances methodologies for IFM baselines by using a causal inference framework to integrate pixel-level

satellite data with a suite of data-driven methods: synthetic control, statistical matching, Random Forest, and Extreme Gradient Boosting (XGBoost). We apply these methodologies to five IFM projects in China to granularly construct ex-post dynamic counterfactuals of carbon stock changes that are likely to have occurred without the project. By comparing this empirical baseline against both the project's self-reported baseline and the observed carbon stock changes, we evaluate the validity of traditional baseline methodologies, the additional impact of project implementation, and the integrity of corresponding carbon credits.

This work has three main contributions. First, we develop dynamic and verifiable baseline scenarios that overcome the limitations in traditional static models by capturing real-world environmental dynamics and reducing self-reported baseline manipulation risks. We demonstrate the effectiveness of multiple data-driven methods for constructing realistic counterfactual scenarios across different forest management contexts. Second, we provide the first empirical assessment of baseline integrity for early IFM projects in China developed under Clean Development Mechanism (CDM) methodologies, which form the foundation for many existing carbon offset standards. This work addresses a research gap by shifting the focus from the predominant study of REDD+ to the comparatively under-explored potential of subtropical and temperate forest management, and geographically, from mature markets like California to high-potential emerging contexts like China.^{32,33} Third, our findings directly inform policy development by providing program operators with evidence-based recommendations for ensuring carbon credit integrity. This research contributes to China's ongoing voluntary carbon market redesign and provides actionable insights for strengthening international forest management

Table 1. Basic Information on Five Forest Management Project Research Samples^a

Project	Province	Methodology	Start time	Year of project design	Expected annual emission reduction (CO ₂ e/yr)	Status
1	Hebei	AR-CM-003-V01 AR-AM0001/version 02	2006	2015	205,090	Registration
2	Yunnan		2015	2016	5,334	Validation
3	Guangxi		2006	2006	25,795	Issuance
4	Sichuan		2007	2009	460,603	Registration
5	Guangxi		2008	2010	87,308	Issuance

^aA project must first undergo “Validation” to be eligible for “Registration”. Only after a project is registered and has generated verified emission reductions can “Issuance” of credits occur.

Table 2. Covariates Used for Counterfactual Baseline Construction: Descriptions and Data Sources

Variable Name	Source	Resolution	Variable Description
Biomass	Liao et al. 2024	500 m	Woody aboveground biomass for China spanning 2001–2020 based on MODIS images, Hansen data set, and GEDI L4B AGB products
NDVI	Yang et al. 2019	30 m	Maximum NDVI for China from 2000 to 2022 based on Landsat remote sensing imagery
Precipitation	Peng 2020	1 km	Global climate data set derived from the National Centre for Atmospheric Sciences of the United Kingdom and the WorldClim global high-resolution climate data set
Solar radiation	He et al. 2020	1 km	Derived from the China Meteorological Forcing Data set, incorporating remote sensing products, reanalysis data sets, and in situ station data
Elevation	Abrams, Crippen, and Fujisada 2020	30 m	Copernicus Global DEM products originating from the European Space Agency
Slope		30 m	Derived from the ASTER GDEM V3 model developed by NASA and JAXA using the ASTER satellite
Soil type	Shi et al. 2024	30 m	Derived from the Second National Soil Survey of China, World Soil Information Service, etc.
Land use type	Xu et al. 2024	30 m	Derived from Landsat remote sensing imagery, the study used data from 2000, 2005, 2010 and 2015.
Forest management type		250 m	Distribution of unmanaged and managed natural regeneration forests, plantation forests with different rotation times, oil palm plantations, and mixed agroforestry, 2001–2020
Distance to road	Made by authors	100 m	Derived from Open Street Map, ArcGIS was used to calculate distance from the study area to the nearest road.
Distance to town	Made by authors	1 km	Derived from LandScan global population density data, ⁴² ArcGIS was used to calculate the distance to nearest populated area (>1500 persons/km ²).

protocols and developing rigorous standards for high-integrity carbon offsets.

METHODS

Analysis Framework

Overview and Framework. The generation of emission reduction credits in offset programs, including IFM projects, typically follows three stages: project design and registration, implementation and monitoring, and verification and issuance (Figure S1). The initial stage mandates Project Design Documents (PDDs) submission, where project developers must quantify anticipated reductions by defining the baseline scenario (the counterfactual state without project intervention) and the project scenario (the state under project activities). This projected impact serves to secure registration and financing. Following successful registration, the project would conduct described project activities and continuously monitor emissions or sequestration changes. The final stage determines the ex-post emission reductions based on third-party verification of the monitored carbon stock against the projected baseline. These ex-ante projections, the ex-post verified credits, and their disparities, constitute core variables and the primary focus of this study. Notably, IFM methodologies calculate emission reductions based on carbon stock changes rather than carbon stock levels, as shown in formulas 1–2.

This study develops a credible baseline identification methodology using ex-post approaches that combine remote sensing data sets with causal inference frameworks. We apply this methodology to five IFM projects in China, comparing estimated baseline carbon stock changes (the annual increment) to observed values during the project period (Figure 1).

The study began with a project selection and data collection. We systematically assessed approximately 150 forestry-based voluntary emission reduction projects registered in China under the Verified Carbon Standard (VCS), CDM, and early CDM-derived methodologies. Projects were included if they met the following criteria: nonzero baseline carbon stock indicating existing forest sequestration

potential, complete geographical boundary data enabling precise spatial analysis, and detailed baseline methodology documentation ensuring methodological comparability. Based on these criteria, ARR projects, VCS projects related to preventing logging, and bamboo forest projects were excluded: preventing logging projects follow REDD+ methodologies that model counterfactual deforestation rates rather than carbon stock growth trajectories, while bamboo exhibits substantially different biological characteristics from tree forests, and the application of tree-based methodologies to bamboo remains methodologically controversial.^{34,35} The final analytical sample comprised five projects: three registered under CDM and two developed using CDM-derived methodologies (Table 1). Notably, CDM reforestation projects on degraded forest land were retained because their baseline scenarios account for pre-existing carbon stocks and natural regeneration potential, making them methodologically analogous to IFM projects.^{36–40}

We extracted key information from project design documents of the five sample projects, including the geographical location, estimated carbon stock for baseline and project scenarios, management activities, and tree parameters. Aboveground biomass (AGB) associated with self-reported carbon stock was then calculated as our core variable of interest. Satellite-based AGB data were collected and processed, with reliability verified through field measurements in selected forest areas where we manually measured AGB within sample plots to provide cross-validation benchmarks for remote sensing data (Section S4 of the Supporting Information).

Using satellite-based data, we constructed baselines through: synthetic controls operated at the project-wide level, optimizing weighted combinations of control groups to minimize characteristic differences between treatment and control groups during pretreatment periods; statistical matching functioned at the pixel level, using covariate matching to ensure control group characteristic distributions closely aligned with project groups before implementation; and machine learning algorithms utilized pixel-level data with XGBoost and Random Forest. The predictive performance of our counterfactual baseline was assessed through placebo tests and error metrics.

Leveraging these well-established baselines, we conducted a 3-fold evaluation. First, to assess the validity of traditional baseline methodologies, we contrasted the projects' self-reported ex-ante baselines with our empirical ex-post estimates. Second, to evaluate project additionality, we compared the observed carbon stocks to the empirical counterfactual trajectories. Finally, to determine the integrity of issued carbon credits, we analyzed the discrepancy between the credits claimed (based on self-reported baselines) and the actual mitigation outcomes (based on the empirical baseline).

Covariate Selection and Control Group Prescreening. To construct a robust counterfactual baseline, we selected ten key covariates to control for confounding factors influencing forest management outcomes. This selection was guided by a systematic review of 24 key studies in related fields (see Table S5), tailored to the specific context of IFM. The covariates encompass: (1) geographical and climatic conditions (annual precipitation, light condition, slope, elevation) as key proxies for agricultural potential and topographical constraints for forest growth; (2) market accessibility and human activity (distance to nearest road and distance to nearest town), which correlate with human activity and market demand and are common proxies for project operational feasibility and harvesting pressure; and (3) ecological and management context (soil organic carbon, NDVI, aboveground biomass, land-use type, forest management type), which are critical for capturing ecosystem health, carbon sequestration potential, and carbon stock trajectories of different management patterns. A complete justification for the variable selection is provided in Section S3 in the Supporting Information, and the data sources are detailed in Table 2.

To ensure the scientific validity and representativeness while balancing computational complexity, we implemented three-step prescreening for the control group selection. First, we restricted potential control areas to the same administrative district as project sites to ensure geographical similarity and to minimize regional bias. Second, areas within 5 km of project sites were excluded to avoid spatial correlation biases, as neighboring areas may experience similar environmental or socioeconomic influences that could compromise spatial independence between the control and treatment groups. Third, we excluded established protected areas based on the World Database on Protected Areas and Chinese Protected Areas to mitigate confounding effects from other conservation measures that could interfere with project effect evaluation.

Following this prescreening process, we employed synthetic control methods, statistical matching, and machine learning algorithms to establish baseline scenarios. Given that the five projects were implemented in different years, we selected the five years preceding each project's implementation year as the pretreatment period for model training.

Data Source

Project Data. Our analytical sample comprised five projects, as detailed in Table 1. Two projects (Projects 1–2) were developed under the methodology AR-CM-003-V01, which was derived from the CDM's afforestation and reforestation framework and used the same tool for identifying baseline scenarios and demonstrating additionality. It was specifically adapted for China by modifying parameters, such as those in growth equations, to better suit local carbon sink projects. The other projects (Projects 3–5) were registered under the CDM's "Reforestation of degraded land" methodology (AR-AM0001). Project boundaries were delineated using ArcGIS and Google Earth based on georeferenced data from PDDs, supplemented by remote sensing imagery where necessary. Projects exhibited variations in the validation status: Project 2 was undergoing validation while the other four were registered, with two CDM projects having issued a certain amount of carbon credits.

A key feature of these methodologies is that the baseline carbon stock change was determined ex-ante in the PDDs and remained fixed throughout the project's crediting period. This baseline carbon stock change ($\Delta C_{baseline_{jt}}$), representing the annual change in forest biomass carbon stocks within the project boundaries in the absence of project

activities, is subsequently converted to the baseline GHG removals by sinks ($GHG_{sink_{baseline_{jt}}}$) using the molecular weight ratio of CO₂ to carbon (44/12), as shown in formula 1. Notably, the baseline accounting exclusively focused on carbon pools in aboveground and belowground woody biomass. Carbon pools in soil organic matter, understory vegetation, deadwood, and litter were omitted, which is deemed conservative as these pools are expected to decrease more (or increase less) in the absence of project activity. Similarly, baseline GHG emissions from fire events and carbon changes in harvested wood products were conservatively excluded. Therefore, the estimation of the annual baseline carbon stock change ($\Delta C_{baseline_{jt}}$) relies fundamentally on the change in total tree biomass ($\Delta Biomass_{jt}$) between time periods t and $t-1$, applying the weighted average carbon content rate of tree species j (CF_j):

$$\begin{aligned} GHG_{sink_{baseline_{jt}}} &= \frac{44}{12} \times (C_{baseline_{jt,t}} - C_{baseline_{jt,t-1}}) \\ &= \frac{44}{12} \times \Delta Biomass_{jt} \times CF_j \end{aligned} \quad (1)$$

The baseline biomass stock $Biomass_{jt}$ at any given year t is the sum of aboveground biomass and belowground biomass. Typically, belowground biomass is derived from AGB using species-specific root-to-shoot ratios (often sourced from IPCC default values). In contrast, AGB is projected using growth models and allometric equations that convert the stand volume into biomass (Table S5). The baseline biomass stock for a given species j in stratum i at year t ($Biomass_{ijt}$) was determined as

$$\begin{aligned} Biomass_{ijt} &= Aboveground_{Biomass_{ijt}} + Belowground_{Biomass_{ijt}} \\ &= Aboveground_{Biomass_{ijt}} \times (1 + R_j) \end{aligned} \quad (2)$$

Therefore, these methodologies effectively make the projected aboveground biomass the most critical determinant for calculating baseline removals. Since the PDDs only reported the final annual baseline GHG removals ($GHG_{sink_{baseline_{jt}}}$), in order to acquire the core biomass data for our analysis, we retroactively calculated the annual baseline aboveground biomass changes, using formulas 1 and 2, and the corresponding parameters (e.g., CF_j and R_j) detailed in each PDD.

Figure S2 presents the ex-ante carbon stock changes reported in the PDDs for both baseline and project scenarios. Projects 1 and 2 exhibit simple baseline trends, while Projects 3–5 adopt highly simplified linear trajectories, indicating static baseline assumptions.

Satellite Data. The core variable of this study, woody aboveground biomass, was quantified using a 500-m resolution annual data set (2001–2020) developed by Liao et al. (2024), which integrates MODIS, GEDI LiDAR, Hansen tree cover, and SRTM topographic data. This data set was selected for its spatial resolution, temporal coverage enabling pre- and postintervention analysis, and calibration for China (see the Supporting Information for details).

Additional covariates utilize various satellite data sets to capture potential impacts of environmental factors and human activities on forest biomass and forest management project development (detailed in Table 2). Figure S3 illustrates selected satellite data maps using the city in which project 1 is located as an example.

Counterfactual Baseline Setting

We designed four data-driven approaches to construct dynamic ex-post counterfactual scenarios serving as baseline scenarios for the five IFM projects. To assess whether these methods detect meaningful carbon stock changes, we applied them to potential control areas with comparable forest characteristics but not subjected to IFM interventions within the same districts during the preintervention period. Since these areas underwent no interventions, observed carbon stock changes are expected to match their counterfactual estimates with minimal errors. For each project, we validated counterfactual methods by comparing estimated versus observed carbon stock within the 5-year pre-intervention period and then

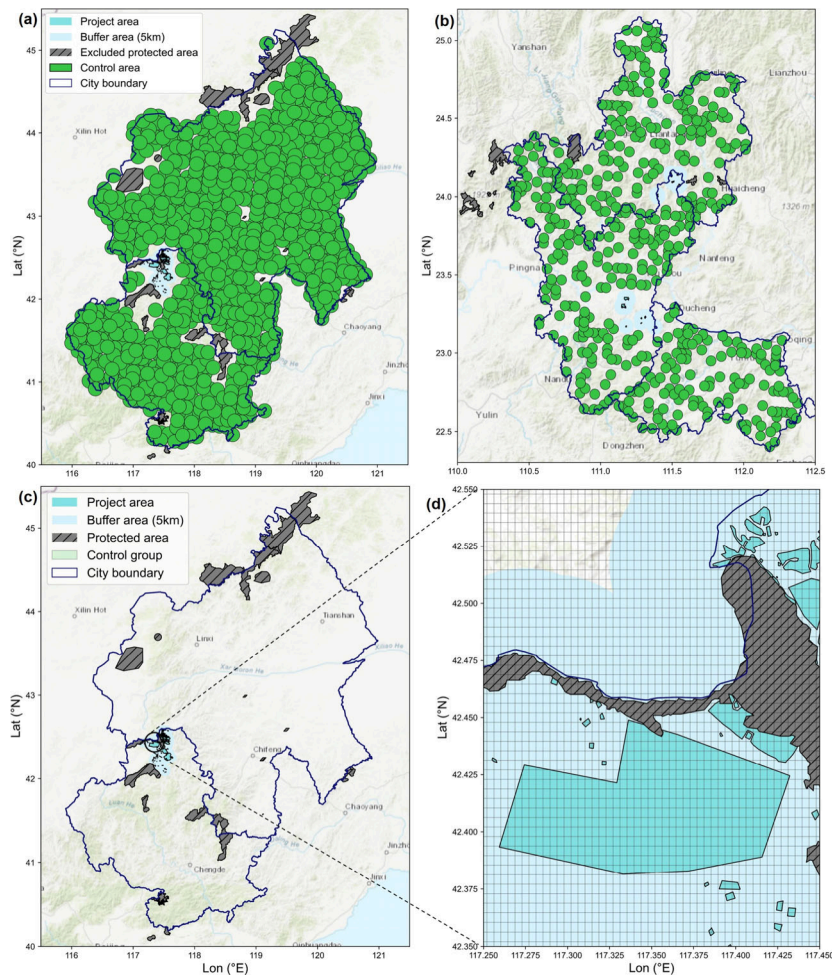


Figure 2. Schematic diagram of the control pool. Plots (a) and (b) present the 1,000 circular spatial polygons created randomly for projects 1 and 3 to construct synthetic control. Plot (c) presents the pixel-based control pool of project 1, and plot (d) is a magnified view of a portion after rasterization.

applied validated methods to construct baseline projections for the post-intervention period.

Synthetic Control Method. The synthetic control method (SCM) constructs counterfactual frameworks using weighted linear combinations of control units,⁴³ which consists of a two-step optimization process to determine optimal weights for potential control units (donors).^{44,45} First, given a variable importance matrix V , weights W are solved to minimize differences between the control's predictor variables (covariates X_0) and treatment group's average predictor variables (X_1):

$$W = \operatorname{argmin}(X_0W - X_1)^T V (X_0W - X_1) \quad (3)$$

$$\text{s.t. } w_i \geq 0; \quad \sum w_i = 1 \quad (4)$$

Second, the optimal weights W^* are determined by the variable importance matrix V such that the synthetic control group's outcome variable trajectory (Z_0W^*) closely approximates the treatment group's outcome trajectory (Z_1) during the preintervention period. This process involves selecting V to minimize the mean squared prediction error (MSPE) between synthetic control's and actual treatment group's outcome values during the pretreatment period

$$V = \operatorname{argmin} \frac{1}{T} [Z_1 - Z_0W^*(V)]^T [Z_1 - Z_0W^*(V)] \quad (5)$$

where T is the number of preintervention time periods, and $W^*(V)$ denotes the optimal weights from formulas 3 and 4 given matrix V . Through this nested optimization, SCM identifies the optimal weights

W that construct a synthetic control closely matching the treatment unit's characteristics and preintervention outcome dynamics, thereby providing a credible counterfactual scenario.

Compared to other counterfactual methods, SCM is particularly suitable for scenarios with few heterogeneous treatment units and limited control groups,^{46–48} which has been extensively applied in evaluating baseline deforestation rates of REDD+ projects.^{19,49} This study adopted a project-level analysis framework, constructing synthetic controls for each of the five sample projects following West et al. methodology.²² For each project, we identified 1,000 potential control units as donor pool candidates. Specifically, these control units were randomly distributed circular polygons located beyond a 5 km radius from project sites within the same prefecture-level city, with each polygon equal in area to its corresponding project site (Figures 2a, 2b and Figure S4).

Statistical Matching. Statistical matching identifies non-intervention units similar to treatment units based on preintervention covariates, simulating counterfactuals and ensuring comparability between treatment and control groups.⁵⁰ This method is extensively employed in environmental and ecological policy evaluation.^{51,52} This study adopts Mahalanobis distance nearest neighbor matching for identifying control units, as it mitigates confounding from inter-variable correlations and scale differences, requires fewer distributional assumptions, and is more robust to outliers.^{53,54} Given a covariate data set $X \in \mathbb{R}^{N \times F}$ with N pixels and F covariates, the Mahalanobis distance between two pixels with covariate vectors $x_1, x_2 \in \mathbb{R}^F$ is calculated as

$$Dis_{Mahalanobis}(x_1, x_2) = \sqrt{(x_1 - x_2)\Sigma^{-1}(x_1 - x_2)^T} \quad (6)$$

where Σ^{-1} is the inverse of the covariance matrix of X .

Unlike SCM's project-level control groups, this study implemented a pixel-level matching strategy (Figures 2c and 2d). The five project areas and their pre-identified control areas were rasterized into 500 m resolution pixels based on satellite-derived aboveground biomass data. This rasterization process yielded 2,672 project pixels and 1,867,582 control pixels, enabling identification of statistically similar controls for project areas (Table 3). The pixel matching procedure involved

two stages. Exact matching was first employed for categorical covariates, specifically, land use types and forest management types. This was followed by one-to-one nearest neighbor Mahalanobis distance matching with replacement based on minimum distance principles, resulting in matches for all project pixels with approximately 2,500 unique control pixels for subsequent analysis.

Machine Learning. Machine learning integration with causal inference has emerged as a significant advancement in economic research, which is widely used for forest classification,⁴¹ land type recognition,²⁵ and aboveground biomass estimation.⁵⁵ This study employed Random Forest and XGBoost algorithms—widely applied in forest research for their robust handling of high-dimensional nonlinear relationships, to construct pixel-based baseline scenarios for IFM projects through a propensity score matching framework (Figures 2c and 2d). We formulated this as a binary classification problem by training models to distinguish project pixels from nonproject pixels based on their observable characteristics, with the generated predicted probabilities serving as propensity scores to identify the most comparable control pixels as counterfactuals. Specifically, pixels from project areas and 70% of prescreened potential control areas served as training data, with the identified

Table 3. Pixel Scale of the Research Areas for Five Forest Management Projects

parameter	value				
Project ID	1	2	3	4	5
Project area	1,783	220	176	102	391
Control area	661,256	133,847	305,855	493,874	272,750

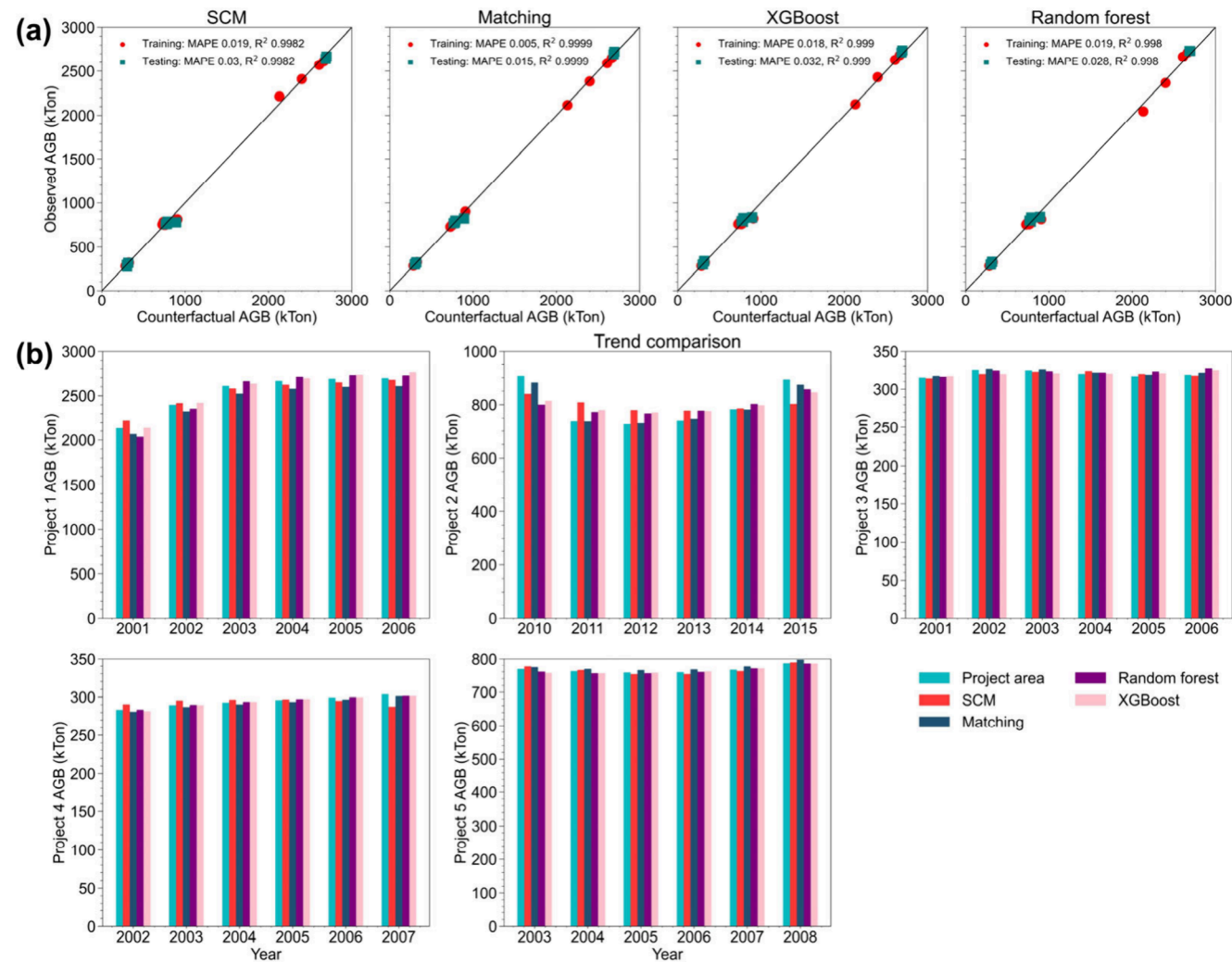


Figure 3. Validation of ex-post baseline estimation approaches. Plot (a) demonstrates comparison of counterfactual versus observed AGB across four ex-post estimation methods. Each plot displays observed AGB on the y-axis against counterfactual AGB on the x-axis, with training set samples shown as red circles and testing set samples shown as green squares. The diagonal line represents a 1:1 relationship. Two performance metrics are reported for each method: MAPE quantifying the deviation between observed and counterfactual AGB and goodness-of-fit (R^2) measuring the coefficient of determination relative to the identity line. Plot (b) presents five-year pre-intervention AGB dynamics by project. Bars compare observed AGB (light blue) in project areas against counterfactual baselines generated by synthetic control (red), statistical matching (dark blue), XGBoost (pink), and random forest (purple).

covariates as predictors and a binary indicator “whether the pixel belongs to the project area” as the outcome variable (project area = 1, nonproject area = 0). The two models were trained and evaluated through 10-fold cross-validation.

The trained classifiers were applied to the remaining 30% of control areas to identify pixels with the most similar forest conditions and environmental characteristics to project areas based on classification probabilities. From these pseudo-project pixels, we selected those whose aboveground biomass closely matched the project area distribution to serve as counterfactual baselines, ensuring comparability in carbon stock levels. Given the imbalanced data set (fewer project pixels than nonproject pixels), we applied the Synthetic Minority Oversampling Technique (SMOTE) to balance training data.⁵⁶ Model hyperparameters were optimized by using random search methods to improve prediction accuracy.

Performance Evaluation and Validation. To ensure the reliability of our counterfactual baselines, we conducted a comprehensive assessment covering predictive validity, covariate balance, and measurement uncertainty. The predictive validity of each method was first assessed during the pre-intervention periods under various timeframes, which were segmented into training and testing phases to verify whether models trained on early historical data could accurately predict the observed subsequent pre-intervention values. Model performance was assessed using Mean Absolute Percentage Error (MAPE) and Root Mean Square Prediction Error (RMSPE), quantifying discrepancies between predicted and observed preintervention values. Covariate balance between treatment and control groups was assessed using the weighted MAPE for synthetic control and the standardized mean difference for other methods, ensuring that differences in means remained within acceptable thresholds relative to the pooled standard deviation (Supporting Information Section S2).

To quantify methodological uncertainty and evaluate the statistical significance of project impacts, we conducted placebo tests following the established permutation testing framework.^{44,57} For each project and each estimation method, we randomly selected 1,000 control areas with identical size to the project sites to serve as pseudo-treatment units. We then applied identical baseline estimation procedures to these placebo units, calculating “placebo effects” as the differences between the observed AGB and counterfactual estimates in the absence of actual intervention. Thus, the distribution of these placebo effects represents each method’s inherent estimation uncertainty, with deviations reflecting natural variability and model fitting errors. Consequently, actual project impacts were considered statistically significant only if they exceeded the 95th percentile of the placebo distribution.

We conducted two complementary analyses: a direct comparison examining whether actual intervention effects substantially deviate from placebo effect distributions, with uncertainty quantified by the standard error of placebo effects across 1,000 trials; and an RMSPE ratio analysis comparing post-intervention to pre-intervention prediction errors, which distinguishes genuine treatment effects from poor model fit persisting throughout both periods. For each unit (actual project or placebo), the ratio is calculated as

$$RMSPE_Ratio_i = \frac{\sqrt{\frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T_{total}} (Y_{it} - \hat{Y}_{it})^2}}{\sqrt{\frac{1}{T_{pre}} \sum_{t=1}^{T_{pre}} (Y_{it} - \hat{Y}_{it})^2}} \quad (7)$$

where T_{pre} and T_{post} denote the number of pre- and post-intervention periods, respectively; Y_{it} denotes the observed AGB for unit i at time t ; and $\hat{Y}_{it}$ is the estimated baseline. We compared the project’s RMSPE ratio against the distribution of ratios derived from the 1,000 placebo units. A project with genuine additionality should exhibit a higher RMSPE ratio than pseudo-treated controls, as treatment effects contribute to post-intervention error. The implied p-value was calculated as the proportion of placebo ratios exceeding the project’s ratio. Statistical significance at the 5% level is achieved if the project’s RMSPE ratio ranks within the top 5% of the placebo distribution.

Additionally, we addressed measurement errors inherent in remote sensing data by adopting a conservative assumption of 20% uncertainty in satellite-derived AGB estimates, consistent with the existing literature and our field validation observations. To construct the final uncertainty bands, we combined the measurement uncertainty with the methodological uncertainty derived from placebo test distributions by calculating their root sum of squares (assuming independence) and subsequently computed the 95% confidence intervals from this combined standard error.

RESULTS

Baseline Scenarios Identification Using Ex-Post Methods

We first assessed predictive validity through temporal displacement testing, which divided the pre-intervention period into training (first 3 years) and testing segments (remaining 2 years plus intervention year), evaluating whether models trained on early data can predict later values (Figure 3a). All methods demonstrated well-calibrated performance with low prediction errors: averaged across projects, synthetic controls achieved training MAPE of 2% and testing MAPE of 3%; XGBoost and Random Forest exhibited excellent generalization with test-set accuracies above 96% and false positive rates below 6%, yielding testing MAPE of 3%, though showing highest RMSPE due to occasional extreme prediction errors; and statistical matching slightly outperformed with mean matched group MAPE below 2%, though individual-pixel MAPE was $\sim 10\%$ due to pixel-level variance. To assess model performance over extended timeframes, we conducted additional validation on Project 2 (initiated in 2015), which offered the longest pre-intervention period. Applying the same training-testing framework yielded consistently robust model generalization (Figure S8). These results confirm that our methods reliably detect meaningful changes and appropriately distinguish real effects from random variation.

Having established predictive validity, we assessed pre-intervention fit performance averaged across the five projects. All methods exhibited strong alignment between estimated counterfactuals and observed project AGB (Figure 3b): synthetic controls maintained strong precision with overall MAPE of 2% and the lowest RMSPE (0.0015); statistical matching showed the most stable performance with mean matched group MAPE of 1% (Table S2); machine learning approaches achieved overall MAPE of 2% with testing-set accuracies exceeding 90% (Table S3). Regarding covariate balance, machine learning methods demonstrated the highest precision with most SMDs within 0.16 and key variables below 0.06; statistical matching also achieved effective alignment though few variables showed slight deviations reflecting trade-offs inherent in multi-variable matching processes (Table S4).

Overall, econometric methods excel in transparent interpretation and parallel trend establishment, while machine learning methods leverage superior high-dimensional data processing for enhanced covariate balance. This complementary strength aligns with Athey and Imbens’ proposition of integrating econometric and machine learning methods for causal inference,⁵⁸ with detailed comparisons in Section S6 of the Supporting Information. The convergence across predictive validity testing, pre-intervention fit assessment, and covariate balance evaluation confirms our counterfactual scenarios reliably capture natural forest dynamics independent of project interventions, providing a robust foundation for baseline setting and additionality assessment.

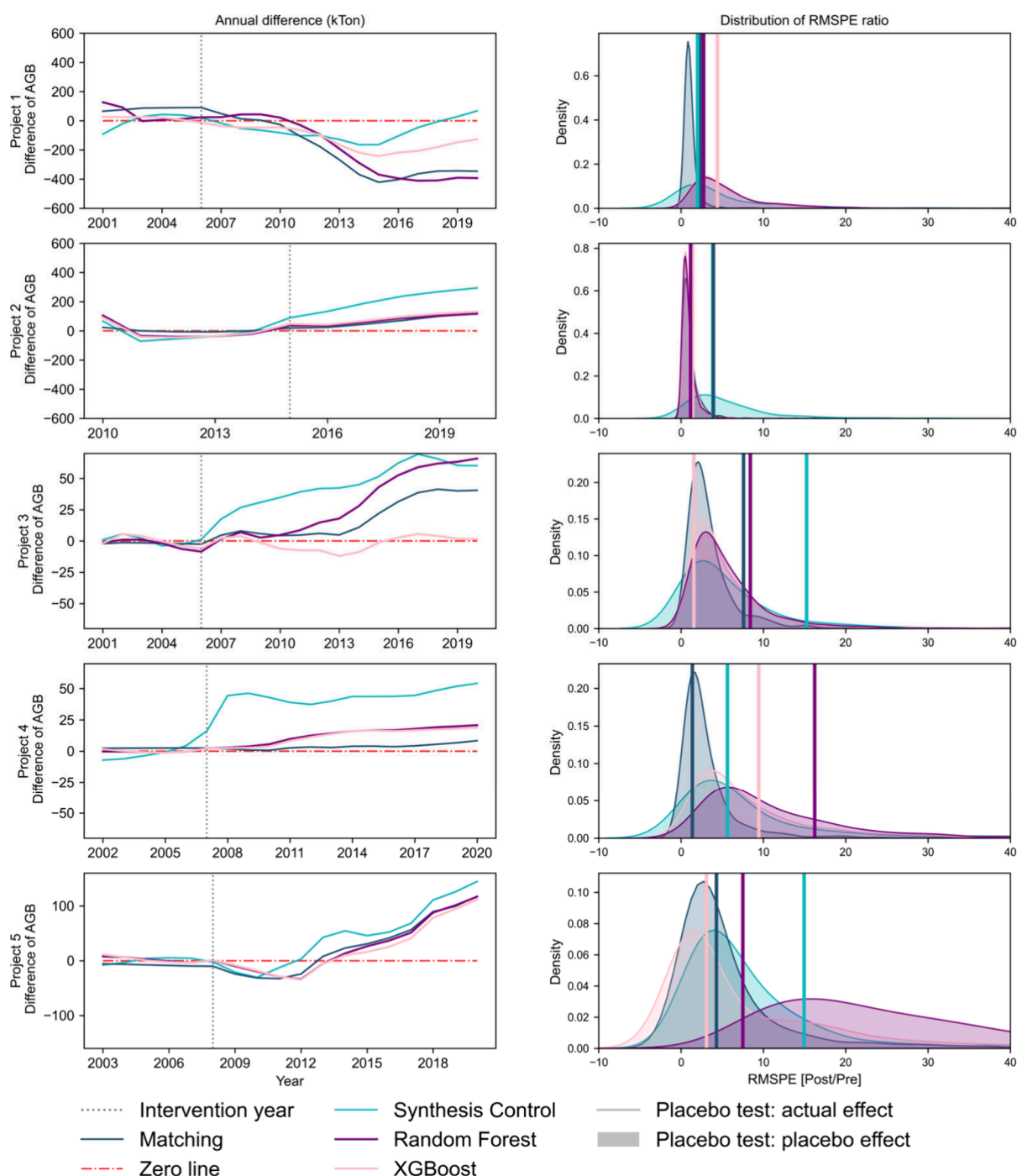


Figure 4. Evaluation of project implementation effects on aboveground biomass. Left panels present estimated intervention effects as annual differences between project and control areas across multiple approaches. Right panels present RMSPE ratios (post-intervention RMSPE relative to pre-intervention RMSPE) for actual projects (vertical lines) against the distribution of RMSPE ratios from 1,000 placebo tests (histograms).

Implementation Effects of the IFM Projects and Uncertainty Analysis

Following the ex-post counterfactual framework, our baseline estimations accurately tracked pre-intervention AGB dynamics across all projects. The differences between observed project AGB and estimated counterfactual baselines (left panels in Figure 4) validate baseline accuracy, remaining near zero across all methods during the pre-intervention period. Modest divergences emerged after intervention, though with substantial variation across methods and projects: synthetic control estimates exhibited the largest deviations, while

statistical matching produced generally more conservative estimates; Project 1 displayed declining trends suggesting potential negative effects, while Projects 2–5 showed delayed impact emergence occurring after 2012 rather than at claimed intervention years, consistent with prior findings of China's voluntary carbon market development.³⁰ Despite these visual patterns, absolute magnitudes remained modest, with annual differences typically below 50,000 tons for Projects 3–5, necessitating further significance tests to distinguish genuine treatment effects from methodological uncertainty and natural variability. We therefore conducted rigorous statistical

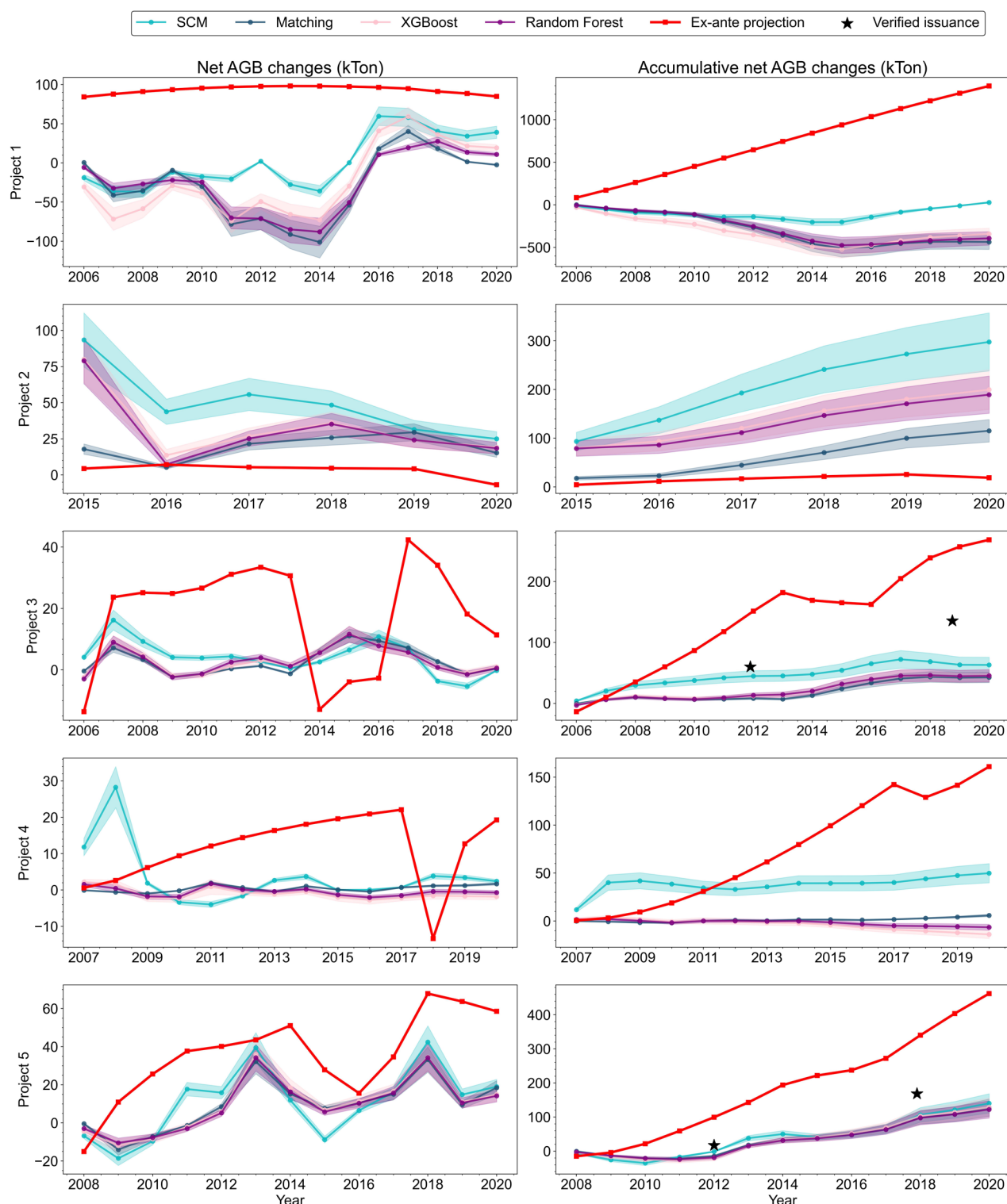


Figure 5. Ex-ante projections, ex-post verified credits, and empirically estimated GHG emission reductions. Left panels indicate annual net AGB changes (annual increments) representing net GHG removal by sinks. Right panels present cumulative net GHG removal by sinks (cumulative storage) over crediting periods, with black stars indicating ex-post verified emission reductions for which credits were issued. Red lines show project ex-ante projections from PDDs, and colored lines show our ex-post empirical estimates from four counterfactual methods and observed data. Shaded areas represent 95% confidence intervals, combining standard errors of the placebo test with 20% measurement uncertainty.

inference through placebo tests and permutation tests by randomly selecting 1000 pseudo-treatment units from each project's control pool and applying identical baseline construction procedures to generate null distributions of placebo effects under the hypothesis of no intervention impact.

Direct comparison of the actual effects against placebo distributions (Figure S6) and the RMSPE ratio distribution (right panels Figure 4) revealed substantial methodological differences. Synthetic control showed wider uncertainty bounds in some cases, reflecting its inherent sensitivity under

convex hull constraints, though estimates remained valid.⁵⁹ Statistical matching produced relatively small and stable standard errors across the projects and periods. Random Forest and XGBoost achieved excellent pre-intervention fit but showed sample size sensitivity, with smaller projects exhibiting amplified post-intervention variance indicative of overfitting; this is further reflected in their RMSPE ratio distributions, where exceptionally small pre-intervention errors inflated the ratios despite moderate post-intervention variance in small projects. For statistical significance assessment, we examined whether actual project effects ranked in the extreme tails (top 5%) of placebo distributions under both direct comparison and RMSPE ratio frameworks. Taking the statistical matching method as an example, Projects 1–5 ranked 1001st, 136th, 652nd, 919th, and 664th among the 1,000 placebo effects for average post-intervention AGB differences, indicating that 100.0%, 13.5%, 65.1%, 91.8%, and 66.3% of randomly selected control areas exhibited stronger post-intervention “effects” than the actual project areas—far exceeding what would be expected if genuine treatment effects existed; correspondingly, these projects showed RMSPE ratios of 2.41, 3.91, 7.58, 1.36, and 4.28, ranking 60th, 52nd, 89th, 690th, and 375th within their respective placebo distributions, none of which exceed the 95th percentile threshold required for statistical significance at the 5% level. Similar patterns emerged across all methods: no project ranked above the 95th percentile simultaneously in both direct comparison and RMSPE ratio tests under any approach, with all implied *p*-values exceeding 0.05.

Based on comprehensive placebo testing across both frameworks and all four methods, we cannot reject the null hypothesis of no significant treatment effect for any of the five IFM projects. These results indicate that observed carbon stock changes in project areas do not significantly exceed what would be expected from natural dynamics and baseline estimation uncertainty, raising important questions about the additionality of claimed emission reductions in these early IFM projects and underscore the necessity of rigorous counterfactual evaluation for carbon credit integrity.

Integrity of Project Self-Reported Baselines and the Associated Carbon Credits

To assess the credibility of project-reported baseline methodologies, we compared self-reported baseline carbon stock changes with our estimated counterfactuals by analyzing their magnitudes and trends. Regarding magnitude, the baseline carbon stock changes for Projects 1 and 2 are reasonably consistent with our counterfactual estimates in absolute terms. In contrast, Projects 3–5, developed under CDM methodologies for reforestation of degraded land, which assume their claimed degraded forestland generates minimal baseline carbon sequestration, reported extremely low annual baseline carbon stock changes: only 13 tons/year (Project 3), 126 tons/year (Project 4), and 656 tons/year (Project 5). Our satellite-based observations estimate substantially higher values of approximately 20,000 tons/year (Project 3), 8000 tons/year (Project 4), and 60,000 tons/year (Project 5). While acknowledging potential systematic differences in carbon accounting methodologies—given that PDDs lacked detailed data sources or monitoring approaches, these extreme discrepancies cannot be explained by methodological differences alone. Instead, they reflect flawed assumptions in the baseline methodology, which inappropriately assume negligible natural carbon sequestration

potential on sites that demonstrate substantial natural growth. Regarding trends, project-reported baselines exhibit oversimplified patterns, with Projects 3–5 adopting simple linear trajectories (Figure S10). In contrast, our dynamic counterfactual estimates capture complex real-world dynamics that are absent from project-reported ex-ante baselines, suggesting that static assumptions in traditional baseline methodologies introduce systematic errors in GHG emissions reduction accounting.

Having identified systematic baseline problems, we assessed their implications for claimed emission reductions by comparing projects' ex-ante projections with our empirical estimates. Project owners reported their projected GHG removals as differences between projected actual carbon stock changes and self-reported baselines, while our estimates use satellite-observed project changes minus our ex-post counterfactual baselines across four methods, shown as annual AGB increment (left panel) and cumulative AGB change over time (right panel) in Figure 5. For Projects 1, 3, 4, and 5, project-reported values (red lines) consistently exceed our estimates (colored lines) and fall outside our uncertainty bounds—combining 95% confidence intervals for methodological uncertainty with 20% measurement uncertainty—in most years. Specifically, Project 1 shows predominantly negative empirical estimates ($-19,050 \pm 20,200$ tons AGB annually), indicating carbon sequestration actually declined rather than increased, with cumulative overestimation reaching 1.68 ± 0.87 million tons by 2020. Project 5 exhibits similar patterns, with claims ($35,540$ tons annually) exceeding our estimates ($9,970 \pm 8,490$ tons annually) by approximately 25,600 tons per year. Project 2 is the only case where empirical carbon sink estimates occasionally exceed ex-ante projections, yet it remains the only project failing to achieve registration—suggesting understated baselines with inflated emissions reduction projections may facilitate project approval. Overall, ex-ante projected removals average 32,200 tons AGB annually across the five projects, exceeding our estimates (5,650 tons annually) by a factor of 3.7 (uncertainty range: 1.7–5.7).

Most critically, Projects 3 and 5 received actual credit issuances based on ex-post monitoring. These ex-post values (black stars in Figure 5 and Figure S7) were verified by third-party verifiers using identical ex-ante baselines and represent cumulative issued credits at each monitoring period's end. The issued credits consistently fall below ex-ante projections, confirming ex-post verification corrects some overestimation, yet remain substantially above our empirical confidence intervals, indicating persistent over-crediting. Project 3 issued approximately 135,400 tons of AGB across two monitoring periods, whereas our 2006–2018 estimate is 51,020 tons (uncertainty range 34,820–82,390 tons, representing the combined range across four counterfactual methods). Project 5 shows even larger discrepancies: the first period (2008–2012) issued ~16,800 tons AGB despite our negative estimate of $-13,960$ tons (uncertainty range: $-25,380$ to -720 tons); cumulative issuances through 2017 reached 168,290 tons AGB against our estimate of 65,420 tons (uncertainty range: 51,070–82,170 tons). Overall, issued credits exceed our empirical estimates by a factor of 1.7 (uncertainty range: 0.8–2.5), demonstrating that while ex-post verification reduces over-crediting compared to ex-ante projections, it fails to correct fundamental baseline inflation.

These analyses reveal systematic integrity problems in IFM carbon crediting practices. Project-reported baselines under-

estimate natural forest carbon sequestration through flawed methodological assumptions and static projections that fail to capture the dynamic forest responses to environmental variability. These deficiencies propagate through the crediting pipeline: inflating ex-ante projections, persisting through verification, and resulting in over-credited issuances. Furthermore, the observed pattern whereby optimistic ex-ante projections coincide with successful registration in our sample suggests potential adverse selection in approval processes, warranting investigation with large samples in the future. Our findings demonstrate that current baseline methodologies and verification procedures are insufficient to ensure the environmental integrity of the IFM carbon credits.

DISCUSSION

Our analysis of five IFM projects in China reveals flawed baseline methodologies and systematic overestimation of emission reductions during both application and issuance phases. These findings align with growing global evidence of overcrediting in forest offsets. Studies examining REDD+ projects have identified inflated baselines as a fundamental challenge, leading to substantial overestimation of carbon benefits from avoided deforestation.^{19,22,60} For IFM projects specifically, research on California's offset program has documented overcrediting of approximately 30% due to methodological flaws in baseline construction,¹⁸ revealed that most forests were accumulating carbon regardless of offset enrollment,¹⁷ and found little evidence of management changes or additionality.⁶¹ Despite employing different methodologies, these findings collectively highlight the urgent need for standardized dynamic baselines for evaluating forest-based interventions. Recent methodological innovations have begun to address this challenge. Takahata et al. (2024) proposed a Bayesian state-space synthetic control framework that unifies ex-ante and ex-post baseline estimation, while Rau et al. (2025) demonstrated biases in ex-ante methods by contrasting one ex-post method using a placebo-based approach, and our research advances this by exploring multiple complementary ex-post dynamic methods in real project contexts.

Our study extends this global evidence base in three critical ways. First, we provide the first empirical assessment of baseline integrity for IFM projects in China, developed under CDM methodologies that have shaped subsequent offset standards globally. Second, we advance methodological applications by integrating four distinct ex-post approaches, whereas previous studies typically employed a single method. This multi-method framework enables explicit assessment of model-dependent counterfactual uncertainty through standardized placebo testing that cannot be obtained from any single approach alone. The consistency of overcrediting patterns across all four methods with different structural assumptions and spatial scales suggests that the fundamental challenge lies in baseline construction paradigms rather than the choice of analytical technique. Third, by demonstrating that these advanced ex-post methods can be applied at fine spatial resolution using publicly available satellite data, our findings support the implementation of rigorous dynamic baseline frameworks at scale. Despite our limited sample size, the consistency of our findings with extensive evidence across different protocols (CDM, VCS, California's offset protocol), ecological zones (tropical, temperate, subtropical), and forest management contexts indicates that underlying challenges

stem from fundamental limitations in current baseline methodologies rather than context-specific failures.

The systematic discrepancies found in this study result from multiple interconnected factors rather than any single cause. Though our study cannot determine the extent to which observed discrepancies result from intentional strategic behavior versus genuine methodological limitations and uncertainties inherent in forest carbon accounting, we present this multi-factorial framework to inspire future research and policy development.

First, baseline setting biases represent a significant contributing factor. Existing IFM protocols typically employ static baseline approaches using historical or regional data and simulation models, which may inadequately capture dynamic forest growth trajectories, with commonly assumed carbon stock growth rates set conservatively low. Our satellite-based causal inference analysis reveals that baseline carbon trajectories exhibit substantial temporal variation driven by climatic variability, natural disturbances, and evolving forest policies—dynamics that static baseline projections fail to capture. The analysis demonstrates that project forests would likely have maintained substantial natural growth rates even without intervention, with carbon accumulation under the ex-post baseline scenarios significantly exceeding project-reported levels.

Second, institutional limitations exacerbate these problems by creating perverse incentives, also highlighted by Haya et al. (2020, 2023). Current IFM methodologies mandate monitoring of project scenario changes during implementation while maintaining static, self-reported baseline values from the application stage. This asymmetric monitoring approach creates information asymmetries where project developers possess detailed forest knowledge, while verifiers rely heavily on project-reported information. Without mandatory ongoing third-party verification for baselines through independent data sources, this structure could inadvertently incentivize project developers to select conservative baseline projections that maximize credit generation.⁶²

Third, project evaluation deficiencies compound these challenges. Project developers typically attribute all net GHG emission reductions to project implementation. However, our permutation tests indicate that the studied IFM projects do not exhibit statistically significant additionality when compared to counterfactuals derived from our four complementary methods. This suggests that observed carbon stock growth may result from multiple factors beyond direct project interventions, including China's national afforestation policies, regional forest recovery processes, and broader environmental trends. Disentangling project-specific effects from these broader dynamics represents a fundamental attribution challenge that requires rigorous causal inference frameworks. Additionally, all five projects make a strong assumption of zero leakage from timber harvest displacement, which may further compound documented overcrediting if displaced timber demand shifts production to other regions. This aligns with concerns raised in existing literature and warrants further investigation in future research.^{30,63}

Based on these findings, we propose four policy recommendations to enhance carbon credit integrity and promote high-quality development of global voluntary carbon markets. First, dynamic adjustment and independent verification mechanisms were introduced for baseline setting. Periodic baseline corrections and mandatory third-party independent

assessments help overcome biases inherent in traditional static baseline approaches with digital MRV systems enabling cost-effective, transparent verification. Second, causal inference techniques and data-driven approaches should be integrated into baseline development and GHG emissions reduction accounting. These methods enhance counterfactual credibility and capture external dynamic factors by selecting suitable controls. Establishing high-quality control area databases for IFM projects would provide scientific foundations for the baseline scenario identification. Third, digital MRV systems, particularly advanced remote sensing technologies, can be leveraged for project monitoring and carbon accounting. These approaches enhance high-frequency, cost-effective, and independent accounting accuracy, while improving market credibility and enabling third-party validation of both project performance and baseline scenarios. Fourth, regular disclosure of monitoring methodologies and data sources can be mandated to enhance transparency. These recommendations address critical challenges in voluntary carbon markets and offer practical solutions for enhancing the integrity, transparency, and reliability of forest-based offsets, thereby strengthening their role in global climate change mitigation efforts.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c11387>.

Supplementary figures and tables, performance metrics, detailed covariate selection, validation of satellite-based biomass data, additional robustness check, and comparative analysis of ex-post baseline methodologies (PDF)

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R.Y. and D.Z. conceived the research; R.Y. and S.Q. acquired data; R.Y. and Z.Z. performed the analysis and created the figures; all authors wrote the paper and contributed to the interpretation of the findings.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We acknowledge support from the National Natural Science Foundation of China (Grant No. 72140005, No. 42341202), the Environmental Defense Fund, and the Tsinghua-Toyota Joint Research Institute Interdisciplinary Program.

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